



Course Name

# ELEMENTARY MATHEMATICS II (CALCULUS)

Topic: Application of Integration to Real-Life  
Problems

WEEK 12

## IN SUMMARY

This week, you learned how integral calculus is applied to solve real-life problems in various fields, including physics, engineering, economics, and environmental science. The lesson covered the practical applications of integration in these domains, showcasing how mathematical concepts can be used to address complex real-world challenges. We explored how integration is used to relate velocity and acceleration in physics, calculate work done by varying forces, analyse electric and magnetic fields in engineering, and much more. Integration was also shown to be a valuable tool in economics for analysing changes in cost, revenue, and profit. Furthermore, you discovered how integration plays a significant role in environmental science, modelling population dynamics, and assessing the cumulative effects of pollutants.



## BULLET POINT SUMMARY

- Application of integral calculus in physics, particularly in calculating distance travelled by an accelerating car and work done by varying forces.
- Integration's role in engineering includes such things as analysing electric and magnetic fields, determining the distribution of forces within structures, and optimising fluid systems.
- Utilisation of integration in economics for marginal analysis, calculating total product and total cost, and assessing market efficiency.
- Environmental science applications include modelling population growth and estimating cumulative pollutant effects in ecological conservation.
- Application of integration in pharmacokinetics for modelling drug concentration over time.
- Integration in medical imaging techniques, like MRI and CT scans, for detailed image reconstruction.

## CASE STUDY

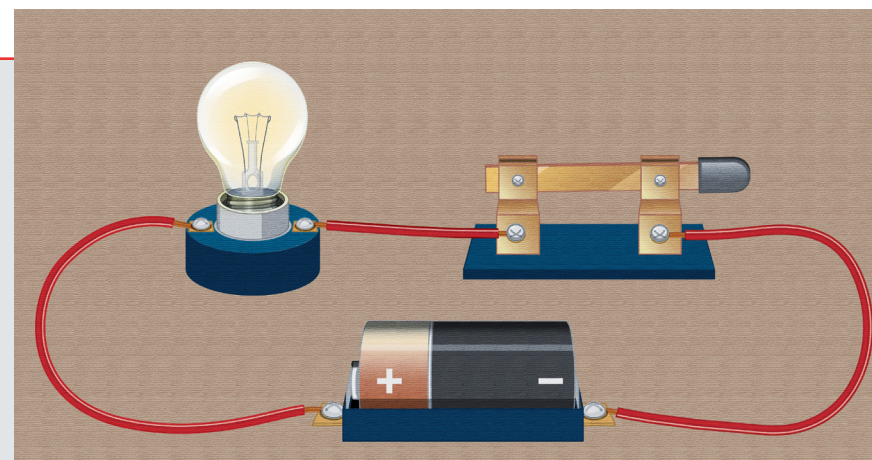
**Title:** The Impact of Integration on Environmental Impact Assessment

**Introduction**

This case study explores how integration is applied in the environmental industry to estimate the cumulative effects of pollutants. It emphasises the significance of this mathematical technique in assessing the environmental impact of various activities.

**Objective**

The objective of this case study is to demonstrate the practical importance of integration in



## CASE STUDY CONT'D

environmental impact assessments. We aim to show how integration helps in quantifying the total volume of pollutants discharged into water bodies over time and its crucial role in ensuring environmental conservation.

**Methodology**

We will consider a scenario in which a factory discharges a pollutant into a river at a varying rate over a specific period. To assess the total pollution over that period, we will employ integration to calculate the integral of the rate of discharge. This will provide a quantitative measure of the environmental impact.

**Results**

The results of this case study will illustrate the concrete application of integration in assessing environmental impact, providing a numerical estimate of the cumulative pollutants discharged during the specified time frame.

## QUESTIONS TO PONDER

- How can integration be further optimised in environmental impact assessments to enhance accuracy and effectiveness in evaluating pollutant effects?
- In what ways can integration-based environmental modelling be used to guide policy decisions and environmental conservation efforts?

**SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON**

After this lesson, you have acquired skills in applying integral calculus to solve real-life problems, which include analytical thinking, mathematical modelling, problem-solving, and the ability to quantify and assess complex real-world issues, particularly in the environmental field.

**PERSONAL REFLECTION**

This lesson has emphasised the wide-ranging applications of integration in various fields, providing a deeper understanding of how mathematical concepts bridge the gap between theory and practical problem-solving. It has underscored the value of mathematical tools in addressing complex real-world challenges, making me more appreciative of the interdisciplinary nature of mathematics.